Eli's Math Study Guide

Simple Solutions Test #5 · Lessons 17–20 · Test day: Friday, Oct 2

How to use this guide

1. Read each skill box. Try the example yourself before you look at the answer.
2. Take the **Practice Test** (25 problems). Show your work!
3. Check your answers with the **Answer Key** on its own page.
4. For anything you missed, go back and re-read that skill box.
5. The **Appendix** has answers to the book problems in Lessons 17–20.

Part 1: Skills

1. Multiplication facts

Lessons 18, 19, 20

How to do it:

• Know your times tables by heart. Order doesn't matter: 7 × 9 is the same as 9 × 7.

• Trick for 9s: hold up 10 fingers and fold down finger number 7. You see 6 fingers on the left and 3 on the right, so 7 × 9 = 63.

Example: 6 × 9 = 54, 7 × 9 = 63, 8 × 7 = 56

2. Factors and factor pairs

Lessons 17, 18, 19, 20

How to do it:

• In 2 × 3 = 6, the numbers you multiply (2 and 3) are **factors**. The answer (6) is the **product**.

• Factors are the whole numbers that divide into a number exactly, with nothing left over.

• To find all the **factor pairs**, start with 1 × the number. Then try 2, 3, 4… Stop when the pairs start to repeat.

Example: Factor pairs for 18: 1 × 18, 2 × 9, 3 × 6 (4 and 5 don't work, and 6 is already used). So the factors of 18 are 1, 2, 3, 6, 9, 18.

Remember: Every number has 1 and itself as factors. Factors of 10: 1, 2, 5, 10.

3. Rounding to the nearest thousand

Lessons 17, 18, 19, 20

How to do it:

• Look at the **thousands** digit, then look at its neighbor to the right (the **hundreds** digit).

• If the hundreds digit is **5 or more**, round **up**. If it is **4 or less**, keep the thousands digit the same.

• Change every digit after the thousands place to 0.

Example: 7,655: the hundreds digit is 6 (5 or more), so round up to **8,000**. 5,262: the hundreds digit is 2, so it rounds to **5,000**.

Remember: “5 or more, raise the score. 4 or less, let it rest.”

4. Comparing and ordering big numbers

Lessons 17, 19, 20

How to do it:

• Line the numbers up by place value. Start at the **left** and compare digit by digit.

• The first place where the digits are different tells you which number is bigger.

• > means “greater than” and < means “less than.” The open mouth eats the bigger number.

Example: 83,407 ___ 83,470. The digits 8, 3 and 4 are the same. In the tens place, 0 < 7, so 83,407 **<** 83,470.

5. Adding and subtracting big numbers

Lessons 17, 18, 19, 20

How to do it:

• Line up the ones, tens, hundreds and so on. Start on the **right** (the ones).

• Adding: if a column adds up to 10 or more, write the ones digit and carry the 1.

• Subtracting: if the top digit is too small, borrow from the next place. With zeros, like 8,000 – 5,379, borrow across the zeros.

• Check subtraction by adding: the answer + the number you took away should equal the top number.

Example: 452 – 197 = **255**. Check: 255 + 197 = 452. ✓

6. Multiplying in parts (distributive property)

Lessons 17, 18, 19

How to do it:

- Break the big number into hundreds, tens and ones (expanded form).
- Multiply **each part** by the small number.
- Add the parts together.

Example: 346 × 4 = (300 + 40 + 6) × 4 → 300×4 = 1,200, 40×4 = 160, 6×4 = 24 → 1,200 + 160 + 24 = **1,384**

7. Division with remainders

Lessons 19, 20

How to do it:

- Ask: “What times the divisor gets as close as possible without going over?”
- The amount left over is the **remainder** (R). It must be smaller than the number you divide by.

Example: 40 ÷ 9: 9 × 4 = 36, and 40 – 36 = 4 left over, so the answer is **4 R4**. 37 ÷ 6: 6 × 6 = 36, 1 left, so **6 R1**.

8. “Times as many” number sentences

Lessons 17, 18, 19, 20

How to do it:

- “Three times as many as 8” means 3 groups of 8. Write it as a multiplication sentence.
- Write the whole sentence with the answer: number × number = answer.

Example: Seven times as many as 4 → **7 × 4 = 28**

9. Place value

Lessons 17, 18

How to do it:

- **Expanded form → standard form:** put each digit in its place. 6,000 + 400 + 20 + 7 = **6,427**.
- Each place is worth **10 times** the place to its right. 10 × 100,000 = 1,000,000.

Example: In 5,550, the 5 in the hundreds place (500) is 10 times the 5 in the tens place (50).

Remember: A digit in one place is worth **10 times** what it is worth in the place to its right.

10. Equivalent fractions and comparing fractions

Lessons 17, 19, 20

How to do it:

• **Equivalent** fractions name the same amount. Draw two bars the same size. Split one into 2 parts and shade 1, then split the other into 6 parts and shade 3. The shaded parts are the same size, so 1/2 = 3/6.

• One half is when the top number is half of the bottom number: 1/2 = 2/4 = 3/6 = 4/8.

• On a number line, the fraction **farther to the right is greater**.

Example: 7/8 is close to 1, and 1/2 is in the middle, so **7/8 > 1/2**.

11. Unit conversions (length and money)

Lessons 17, 18, 19, 20

How to do it:

• Going from a **bigger** unit to a **smaller** unit (yards → feet → inches)? **Multiply.**

• Going from a **smaller** unit to a **bigger** unit (inches → feet → yards)? **Divide.**

Example: 7 yards = 7 × 3 = **21 feet**. 36 inches = **1 yard**. $8.00 = 8 × 4 = **32 quarters**.

Remember: 1 foot = 12 inches · 1 yard = 3 feet · 1 yard = 36 inches · $1.00 = 4 quarters

12. Geometry words and perimeter

Lessons 17, 18, 19, 20

How to do it:

• **Point:** a dot with a capital-letter name (point B). **Line:** goes on forever both ways (arrows on both ends).

• **Line segment** BC: has two endpoints. **Ray** ST: starts at endpoint S and goes forever through T (one arrow).

• **Vertical** line: straight up and down. (Horizontal is side to side.)

• **Perimeter** = the distance all the way around a shape. Add up all the sides.

Example: A triangle has 3 sides that are each 5 cm. Perimeter = 5 + 5 + 5 = **15 cm**.

13. Word problems

Lessons 17, 18, 19, 20

How to do it:

• **“That is ___ more than…”** problems: the big number is given, and you need the smaller one, so **subtract**. “The zoo bought 93 lb, which is 29 more than last month” → 93 – 29 = 64 lb.

• **Choose the equation:** pick the one that subtracts the “more” amount from the bigger number.

• **More or fewer than…?** Round each number to its highest place value, add them, and compare. If the rounded total is **exactly** the target, add the real numbers to be sure.

• Always write the **unit** in your answer (feet, stickers, degrees…).

Example: 321 + 251 + 111 + 201 → 300 + 300 + 100 + 200 = 900. 900 is more than 800, so she filed **more** (the exact total is 884).

Part 2: Practice Test

Name: _____ Date: _____ Score: _____ / 25

Show your work in the space under each problem. For multiple choice, circle the letter.

1. 8 × 9 = _____

2. In 4 × 7 = 28, which numbers are the **factors**, and which number is the **product**?

3. List all the factor pairs for 12.

4. List all the factor pairs for 20.

5. Round 3,509 to the nearest thousand.
a) 3,000 b) 3,500 c) 4,000 d) 3,510

6. Write <, >, or = : 52,614 ____ 52,641

7. Put these numbers in order from greatest to least: 7,425 7,452 7,405

8. 64,739 + 28,586 = _____

9. 6,000 – 2,847 = _____

10. Use the distributive property to multiply 238 × 4. Multiply 4 by all the values in 238 (200 + 30 + 8). Show the product.

11. Use the distributive property: 615 × 3 = ? (600 + 10 + 5)

12. 29 ÷ 4 = _____ (There will be a remainder.)

13. 50 ÷ 8 = _____ (There will be a remainder.)

14. Write and solve a number sentence for five times as many as 6.

15. Write 4,000 + 700 + 50 + 3 as a base-ten number.

16. In the number 5,550, the 5 in the hundreds place is worth _____ times the 5 in the tens place.

17. If 1 yd = 3 ft, then 5 yd = _____ ft.

18. Ben’s kite string is 2 yards long. How many **inches** long is it?

19. Fill in the blank so the fractions are equivalent: 1/2 = ____/6. Then draw a fraction model to show it.

20. Write <, >, or = : 5/8 ____ 1/2 (Hint: think of a number line.)

21. Maria has 52 marbles. That is 18 more than Sam has. How many marbles does Sam have? Choose the correct equation and solve it.
a) 52 + 18 = ? b) 52 – 18 = ? c) 18 – 52 = ?

22. A bakery sold 412 muffins on Monday, 189 on Tuesday, 305 on Wednesday, and 96 on Thursday. Did they sell more or fewer than 900 muffins? (Round to the highest place value.)

23. Jake needs $5.00 in quarters for the arcade. He has 7 quarters. How many more quarters does he need?

24. Which one has **one endpoint** and goes on forever in one direction?
a) line segment b) ray c) line d) point

25. A square has sides that are each 6 inches long. What is its perimeter?

Practice Test: Answer Key

For a grown-up: check one problem at a time and talk through any mistakes.

1. 72

2. Factors: 4 and 7. Product: 28.

3. 1 × 12, 2 × 6, 3 × 4

4. 1 × 20, 2 × 10, 4 × 5

5. c) 4,000 (the hundreds digit is 5, so round up)

6. 52,614 **<** 52,641 (tens place: 1 < 4)

7. 7,452; 7,425; 7,405

8. 93,325

9. 3,153 (check: 3,153 + 2,847 = 6,000)

10. 800 + 120 + 32 = **952**

11. 1,800 + 30 + 15 = **1,845**

12. 7 R1 (4 × 7 = 28, 1 left over)

13. 6 R2 (8 × 6 = 48, 2 left over)

14. 5 × 6 = 30

15. 4,753

16. 10 (500 is 10 × 50)

17. 15 ft (5 × 3)

18. 72 inches (2 × 36)

19. 1/2 = **3**/6. Model: a bar split into 2 with 1 part shaded, and a bar of the same size split into 6 with 3 parts shaded.

20. 5/8 **>** 1/2 (1/2 = 4/8, and 5/8 is farther right)

21. b) 52 – 18 = 34 marbles

22. Rounded: 400 + 200 + 300 + 100 = 1,000, so **more** than 900 (the exact total is 1,002).

23. $5.00 = 20 quarters, and 20 – 7 = **13 quarters**

24. b) ray

25. 6 + 6 + 6 + 6 = **24 inches**

Appendix: Answers to the Book Problems (Lessons 17–20)

These answers were worked out for this guide. They are not printed in the book. “Drawing” means Eli draws it in the book. Two problems need the picture in the book.

Lesson #17
1. Factors 3 and 9, product 27
2. 5,000
3. 1,200 + 160 + 24 = 1,384
4. Drawing (1/2 = 3/6)
5. 158,543
6. 1, 2, 5, 10
7. 9,681; 9,618; 9,600
8. 255
9. Drawing (up-and-down line)
10. 93 – 29 = 64 lb
11. More: 300+300+100+200 = 900 (exact 884)
12. 3 × 8 = 24
13. 72 in
14. 6,427
15. Drawing (a dot labeled B)

Lesson #18
1. 63
2. Drawing (ray from S through T)
3. Needs the picture in the book
4. 1×18, 2×9, 3×6
5. 800 + 320 + 28 = 1,148
6. 10
7. 1,422
8. 101 – 2 = 99 degrees
9. 6 in (12 ÷ 2)
10. 1×24, 2×12, 3×8, 4×6
11. 4,000
12. 1,086
13. 1 yd
14. 7 × 4 = 28
15. More: exact total 895 (rounding gives exactly 800, so add the real numbers)

Lesson #19
1. 48 in
2. 4 R4
3. 54
4. Drawing (segment B to C)
5. <
6. 1×16, 2×8, 4×4
7. Needs the picture in the book
8. 8,000
9. 160,273
10. 23 – 17 = 6 stickers
11. 21 ft
12. 2,621
13. 2,800 + 200 + 8 = 3,008
14. 7/8 > 1/2
15. 4 × 3 = 12

Lesson #20
1. 3,260
2. 8 × 10 = 80
3. 15 – 9 = 6 home runs
4. 27,465
5. 1×24, 2×12, 3×8, 4×6
6. 6 R1
7. Drawing (1/2 = 4/8)
8. Drawing (segment J to K)
9. 9 ft
10. 8,000
11. 56
12. 24 yd = 72 ft
13. <
14. 32 – 9 = 23 quarters
15. 81,489